

1- Face to Face Interaction

Face to face interaction exists among students when students:

- (1) orally explain to each other how to solve problems,
- (2) discuss with each other the nature of the concepts and strategies being learned,
- (3) teach their knowledge to classmates, and
- (4) explain to each other the connections between present and past learning.

This face to face interaction is promotive in the sense that students help, assist, encourage , and support each other's efforts to learn.

2- Individual Accountability

Individual accountability/Personal Responsibility requires the instructor to ensure that the performance of each individual student is assessed and the results given back to the group and the individual.

The group needs to know who needs more assistance in completing the assignments and group members need to know they cannot "hitch-hike" on the work of others.

Common ways to structure individual accountability include:

- (1) giving an individual exam to each student,
 - (2) randomly calling on individual students to present their group's answer,
 - (3) giving an individual oral exam while monitoring group work (e.g., while the individual student is delivering an 'expert' tutorial to a small group).
-

3- Group Processing

Group processing (e.g., process check) involves a group discussion of how well they are achieving their goals and how well they are maintaining effective working relationships among members.

At the end of their working period the groups process their functioning by answering two questions:

- (1) What is something each member did that was helpful for the group and
- (2) What is something each member could do to make the group even better tomorrow?

Such processing: (1) enables learning groups to focus on group maintenance, (2) facilitates the learning of collaborative skills, (3) ensures that members receive feedback on their participation, and (4) reminds students to practice collaborative skills consistently.

4- Social Skills

- **Social (collaborative) skills are necessary for effective group functioning.**
- **Students must have and use the needed leadership , decision-making, trust-building, communication, and conflict-management skills.**
- **These skills have to be taught just as purposefully and precisely as academic skills.**
- **Many students have never worked cooperatively in learning situations and, therefore, lack the needed social skills for doing so.**

5- Positive Interdependence

- **Positive Interdependence exists when students believe that they are linked with others in a way that one cannot succeed unless the other members of the group succeed (and vice versa).**
- **Students are working together to get the job done. In other words, students must perceive that they “sink or swim together.”**
- **In a problem-solving session, positive interdependence is structured by group members**
 - (1) agreeing on the answer and solution strategies for each problem (goal interdependence) and**
 - (2) fulfilling assigned role responsibilities (role interdependence).**
- **Other ways of structuring positive interdependence include having common rewards, being dependent on each other's resources, or a division of labor.**

The Assignment's Learning Objectives

The learning objectives for this assignment are to help you:

1. To achieve Awareness about several fields of engineering, and
2. introduce you to how to present technical work, and
3. To achieve Comprehension in your ability to research and present a topic.

The Assignment

You are to search and become an “expert” about **certain** engineering major. Once you have become an “expert” you are to prepare a **five**-minute oral tutorial about this major that you will present to a small group of your colleagues in class.

There are numerous things you could include in your tutorial; a few possible examples are:

1. What sort of problems do engineers in your field typically work on?
2. What is the typical project names and durations?
3. What sort of companies hire engineers in your field?
4. How many engineers are there in your field (local, national or international),
5. What is the name of the professional society in your field?
6. How the future of this profession looks like, or what does the job market (currently and in the coming 20 years) looks like for engineers in your field,
7. What is/are the engineering professions that this profession stemmed from,
and
8. What are the engineering professions that may be branched from this profession?
9. Additional points are permitted

There are many places you can go to get information (e.g., the library, the Web, etc.). Many of the professional societies have web pages at "www.profsoc.org", where "profsoc" is the name of the society. A web search (e.g.; google ...+profession).

Your instructor will assign one specialization for each student to work on it. The following list may be used for this purpose:

Electrical Engineering: power engineering, electronic engineering, power electronics engineering, communications engineering, computer engineering, automatic control engineering and industrial automation, signal engineering and more.

Civil Engineering: structural engineering, geotechnical engineering, environmental engineering, transportation engineering, water resources engineering, earthquake engineering, materials engineering, building engineering, surveying engineering, municipal engineering, coastal engineering, tunnel engineering, Sanitary Engineering and more.

Mechanical Engineering: Micro Electromechanical Systems (MEMS) engineering, material engineering, Mechatronics engineering, Biomechanics engineering, Robotics engineering, control engineering, Construction and Structure engineering, space engineering, Design and Production Engineering, Automobile engineering, marine engineering, Aviation Engineering and more.

Chemical engineering: Materials Processing, Materials Synthesis, Environmental Engineering, Alternative Energies, Biomedical Chemical Engineering, Food Engineering and more.

Petroleum Engineering: Drilling Engineering, Stock Engineering, Petroleum Production Engineering and more.

Industrial Engineering: industrial engineering, manufacturing engineering, quality engineering, process engineering and more

Misc: mining engineering, nuclear engineering, fire-fighting engineering, bio engineering, bio-instrumentation engineering, biomaterials engineering, clinical engineering and medical imaging engineering.

The Submission

- This assignment, totally, must be word processed in English
- You will submit the material you used in preparing your tutorial to the associated link on the BB prior to the associated concept session. This material is called **Tutorial materials**. It must be presented in less than **four** pages (pages not papers).
- However; the checklist, Tutorial itself and any other work are to be submitted in hard copies. The top sheet of your submission should be the assignment checklist. Although it is not required to add the tutorial materials here; it will not affect the assessment if it was to this package.

The Assessment

Your tutorial and tutorial materials will be assessed using **Engineering the Profession Tutorial Materials Checklist**

Engineering the Profession **Tutorial & Tutorial Materials** Checklist (2026) PI: 7.1

Name Group Team Member

Work Evaluated by (inst/grader) _____ Assessment Symbol/Color _____ Date ____ / ____ /202...

Yes	No	Self Regulation Issues (A lapse will be issued for each No in this section)
		A. The tutorial materials was submitted (electronically) to the Blackboard prior to the due time in MS-word format
		B. The tutorial materials is in less than four pages (pages not papers)
		C. The top sheet of the submitted hardcopy-assignment is this checklist with only (the top row & the “LOCATION” column in the expected features) are completed by the student before submission.
		D. No cover sheet is added
		E. The TUTORIAL is one page only
		F. The TUTORIAL is located just after this page. The PTW comes next
		G. There are extra materials by the end and not subject of assessment (type NA if not exist)
		Enter the sum of the No(s)

Yes	No	Expected Features	
		1. The tutorial materials were assessed as at least NI (in the PTW)	
		2. It is clear which engineering discipline is being discussed.	
		3. The material is primarily related to the profession’s information (not about a college program).	
		4. The assignment is Word processed and in English.	
		ITEM	Section (student must state this)
		5. the sort of problems do engineers in this field typically work on is clear	
		6. It is clear what is the typical project names and durations	
		7. It is clear what sort of companies hire engineers in this field	
		8. There is an approximate number for the engineers in this field	
		9. The name of the professional society in this field is stated in	
		10. It is clear what the future of this profession looks like	
		11. The fathers and the branches of this filed are clearly stated	

Discuss any No’s for Expected Features

RESULTS and ASSESSMENT	
M	All yes in the expected features and no more than ONE “NO” in the self-regulations section
NI	The total number of “NO” is less than 4
NCE	Any other combination
NS	no tutorial materials were submitted
Enter total SR-Lapse (lapses equal or more than the number of “Nos” in the SR-section)	

Students: Correction for this assignment is not permitted

For instructor use only: the percentage of the plagiarism is acceptable	yes	No
---	-----	----

Original quiz Checklist (481)

Name: _____

Quiz Date: / /2026

Quiz Number:

1	2	3	4	5	6	7	8	9	10
---	---	---	---	---	---	---	---	---	----

group	
-------	--

	Team number
--	-------------

	member number
--	---------------

Quiz Reading Assignment: _____

1. Work Evaluated by _____ Assessment Symbol/Color _____ Date _____

2. Work Evaluated by _____ Assessment Symbol/Color _____ Date _____

You are to complete **only the top 4 lines** of this page. The grader will complete this checklist

Yes	No	Self-Regulation Issues (completed by instructor &/or grader) – half lapse for each No
		A. The quiz was submitted on time (if No, instructor initial under No)
		B. This checklist is added by the student not by the instructor & is located on the top of the quiz-sheet with the correct orientation
		C. This checklist is stapled from the left to the quiz-sheet
		D. Only the top 4 lines of this page are completed by the student prior to the submission and correct.
		E. There were no other Self-Regulation Problems (describe if No _____)
		Enter the sum of the No(s)

Yes	No	Summary of Journal Performances (if this is a makeup)
		An engineering journal was submitted
		If journal submitted, the engineering journal met expectations (see Journal Checklist)

Summary & Results

Total number of No in the self-regulation table			
Total number of Lapses			
%	Without Accepted Journal	With Accepted Journal	
100	E	E (+Bonus)	
90	M	E	
80	M	M (+Bonus)	
60-70	NI	M	
40-50	NI	NI	
0-30	NCE	NCE	
other	NS	NS	

Please note that: No “E” for more than zero lapses whatever the situations

Engineering Journal sub-Checklist (2026)

Student Name

Quiz Date:

Quiz Number

Group Number

Team Number

Member Number

Quiz Name

1. Work Evaluated by _____ Assessment Symbol/Color _____ Date _____

2. Work Evaluated by _____ Assessment Symbol/Color _____ Date _____

You must **self assess**, i.e., fill out this checklist, **attach** it as the **cover sheet** to your Engineering Journal, and place it into the Team Folder **before** beginning the associated Quiz.

Yes	No	Expected Feature
		1. This completely filled out and self-assessed checklist is stapled to the front of the Engineering Journal Entry and correct .
		2. The PTW-checklist is the first page after this page.
		3. The evaluation on the PTW is "M".
		4. The Journal is handwritten and two pages or less.
		5. The top of the journal includes the student's Name , Team Number & Member Number as well as the Quiz Date & Name on each page.
		6. The work is professional, ethical and is presented using the Presentation Sandwich .
		7. The journal writing covers all the parts of the reading/required materials
		8. The writing is truly reflective and contains two or more of the following (check if present):
		• a summary , in the student's own words, of the important concepts in the material and why they are important;
		• a list of questions about concepts or a description of why items in the reading were unclear to you;
		• a personal description of potential applications of the material to the student's own life (written in the first person);
		• comparison and contrast of the reading material to other material in this or other courses; and
		• other clearly reflective writing: name it (.....).
Comments (if any)		

Results of the Assessment

M	meets expectations, Yes for all items of the expected features
NI	needs improvement, only one No in the expected features
NCE	no credible effort , more than one NO; this reflect that this work does not demonstrate a credible effort to meet the specifications summarized above.

For student use

STUDENT PERFORMANCE SUMMARY in CONCEPT/LAB

Student ID:

Name:

Team:

member:

[illegible]

For final grade: compare the number of exceeds versus the number of Number of defects

العالم الخاص

الأفكار الخاطئة التي تعيق التعلم

يبدأ العرض بمشاهد للحفلة السنوية لخريجي جامعة هارفرد حيث تم انتقاء 23 شخص بشكل عشوائي من منسوبي وطلاب الجامعة لإجراء دراسة على مدى وجود أفكار خاطئة لدى الناس، وقد تم ذلك عن طريق طرح ثلاثة أسئلة هي:

1. ما هو سبب حدوث الفصول الأربعة؟
2. ما هو سبب تغير أوجه القمر خلال الشهر؟
3. ما عدد مواد العلوم الطبيعة التي تم دراستها خلال مسيرته العلمية؟

وقد أتضح من تلك الدراسة أنه بغض النظر عن عدد المواد المتعلقة بالعلوم الطبيعة التي تم دراستها لوحظ أن 21 شخص من 23 موضع الدراسة لديهم أفكار خاطئة عن أسباب حدوث الفصول الأربعة وسبب تغير أوجه القمر، وقد تمثلت بعض تلك الأفكار في أن اختلاف المسافة بين الشمس والأرض أثناء دورانها حولها هو سبب حدوث الفصول الأربعة وأن اختلاف انعكاس ضوء الشمس على القمر هو سبب اختلاف أوجه القمر.

و كذلك من أجل اختبار مدى نجاح أو فشل التوجيه السليم وطريقة التعليم في إزالة تلك الأفكار الخاطئة لدى الطلاب تم اختيار 9 طلاب من الثانوية العامة (ثانوية كامبردج ريندج والثانوية اللاتينية) وتم طرح السؤالين الخاصين بسبب حدوث الفصول الأربعة و تغير أوجه القمر، وقد تعددت إجابات الطلاب الخاطئة بين اختلاف المسافة بين الشمس والأرض هو سبب حدوث الفصول الأربعة وحتى أن الغيوم هي سبب اختلاف أوجه القمر!!!

وبرينا العرض مدى اندهاش المعلمة بإجابات طلابها المختلفة وعن كمية الأفكار الخاطئة التي لديهم. وحتى عينة الاختبار التي قيمتها المعلمة بمستوى 9 من 10 والتي كانت تتوقع أنها ستجيب على الأسئلة بشكل صحيح كان لديها عدة مفاهيم وأفكار خاطئة تتمثل فيما يلي:

◀ أن الأرض تدور حول الشمس في مدار غير منتظم.

◀ إمكانية انحناء أشعة الشمس وهذا هو سبب حدوث الفصول الأربعة.

◀ أن سبب تغير أوجه القمر هو ظل الأرض.

و يتضح من ذلك أن لكثير من الطلاب أفكار خاطئة تتعارض مع أبسط المبادئ العلمية كفكرة انحناء أشعة الضوء.

وعلى ضوء تلك النتائج تم إعطاء الطلاب دورة حول علم الفلك فيما يختص بدوران الأرض حول الشمس ودوران القمر حول الأرض باستخدام أدوات شرح مساعدة مثل النماذج المصغرة وذلك لمدة أسبوعان تم بعدها مقابلة العينة نفسها مرة أخرى وإعادة طرح الأسئلة ذاتها لوحظ وجود تغير كبير في فهم الموضوع وتغير ملحوظ في كمية الأفكار الخاطئة لديهم، كما لوحظ زوال الارتباك القائم حول بعض المفاهيم عند استخدام الوسائل المساعدة.

وبالرغم من ذلك فإن بعض الأفكار الخاطئة مازالت عالقة في الأذهان لدى عينة الاختبار الأخيرة كالفكرة عن الضوء المباشر وغير المباشر وهي تمنعها من تقبل مفهوم تأثير ميلان محور الأرض على حدوث الفصول الأربعة.

وفي الختام يتضح لنا أن وجود أفكار أو مفاهيم خاطئة لدى الإنسان تعيقه من تقبل الأفكار السليمة والتعامل معها بشكل جيد، لذا علينا العمل على التخلص من تلك المفاهيم الخاطئة وتصحيحها لنتمكن من الاستمرار في التعلم والإبداع.